

But what if there were just one passenger in the bus? Such scenarios are harder puzzles to solve.

Aside from matters regarding interpretation of language, there is also the issue of conflicts between the laws themselves. For instance, the first and second law could come into conflict in some cases. Like, what if a physician robot knows that a patient who is currently unconscious would harm other people when he awakes? Should the robot not treat the patient?

A conflict also arises whenever a human orders a robot to protect him or her from another human who intends to kill them. If restraint is not a possibility, what route would the robot take?

While self-driving cars are all the talk of the town, we can look at a potential scenario involving a robotic driver. Say, the robot is driving and its human owner is the passenger. To avoid an accident, would the robot swerve off a cliff that would cause the death of the passenger, or do nothing, which could lead to the death of many more people on the road? Going by the logic of numbers, the robot should sacrifice the life of one for saving many. However, if you recall, the passenger is the robot's owner, and isn't it the robot's primary directive to protect its owner's life?

Such dilemmas are just the tip of the iceberg as far as conflicting scenarios involving the three laws of robotics are concerned.

The alternatives

While Asimov's three laws of robotics offer a useful springboard in matters regarding machine ethics, they are by no means the only possibilities that can be explored. Indeed, many people – very smart people – are exploring other models of ethics and rules that can be used in robotics.

Take for instance the case of the commercial toy robot named Nao. The robot got programmed so that it could remind people to take their medicine on time. A simple enough proposition. However, even in this simple scenario, there were ethical challenges. For instance, what if the patient refuses to take the medication? If the robot allows the patient to skip the medicine, it could be



How to train your robot in ethics?